

Tapping In - How to Decide: Mind, Heart, or Gut?

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ABSTRACT

Motivation: Immediate cognition without rationalization is called intuition - an empowering faculty. Many of us feel disconnected from our intuition, potentially causing us to struggle with confidently making decisions. **Challenge:** We set out to find out why that is and how to support aligned decision-making. **Method:** We interviewed fourteen experts on tuning into our intuition for decision-making, surveyed currently available decision-making support tools, and modeled a tool that extends current approaches with the expert insights. **Results:** Based on the experts' insights, we modeled a decision trifecta including mind, heart, and gut along with a narrative of how to educate on their interplay and how to use that to take more aligned decisions. We critically question to what extent decision support technology is a beneficial way forward. **Impact:** The model opens up a space for discussion around holistic decision making from an individual's perspective. It serves as a reflection tool for personal processes as well as the suitability and limits of supportive technology.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; *Heuristic evaluations*; • **Applied computing** → *Psychology*.

KEYWORDS

intuition, decision, decision-making, decision support, analysis

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1 INTRODUCTION

Have you ever mulled over a decision for a long period of time? Have you endured the agony of endless plus and minus lists? Have you ever engaged in unproductive rumination circles around a situation in your life? We have experienced a fair share of that, and maybe you can relate. When facing difficult decisions, such as changing a career path, furthering or ending a significant relationship, or moving to a new geographic location, we often find

ourselves focused on rational and emotional decision making. Our intuition, or gut instinct, is also a reliable indicator, but we often miss or dismiss it. When we ignore our intuition, our emotional and rational narratives may conflict and we may enter decision loops. We will bounce back and forth between listening to what our heart tells us versus our mind. For an example of ignoring intuition in decision making and getting stuck in decision loops, we present the story of a persona's career change: Kim has been a software developer with a large firm for the last seven years. The salary is good, the benefits are attractive, the long-running contract feels safe, but the long working hours are excruciating. Kim feels like creativity has been sucked out of them. Kim always had a knack for illustrating. They like to draw and design interfaces in their free time. They dream about becoming a freelancer, but are afraid of the uncertainties that come with such a switch. Kim feels pulled in the opposite direction when they acknowledge the good that could come from making a change. They continuously question whether they should "follow their dream" or stick with the safety they currently have. They feel uncomfortable in the thought loop and cope with distraction and postponement. If Kim were to ask they would get three general pieces of well-meant advice: "make a pro/con list and think it through," "follow your heart," and "go with your gut." This advice shows the three main decision engines we can perceive in our body, the mind (rationale), the heart (emotions), and the gut (intuition). Although we may think we are predominately using our brain to make decisions, we are not [61]. Decisions are formed in our body and then our rational minds build the necessary narrative to go with it [49]. One solution for taking hard-to-make decisions is to triangulate decisions using intuition, emotions, and rationale. In Michael Singer's words, we may not think that we do, but inherently by nature "you already know how to find yourself; you have just gotten distracted and disconnected" [56, Intr.]. In other words, we have become disconnected from our intuition. **The underlying premise of this paper is that the process of decision making often needs to take intuition into consideration instead of relying on purely rational, and in some cases emotional, decision making.** A similar assertion was made in 1999 in context of decision-support systems research and development. In a CACM paper, Vicky Sauter stated "intuition [was] becoming increasingly important in decision-making" and suggested that decision support systems (DSS) should incorporate ways to allow for testing intuitive hypotheses [50]. In recent years intuition-based decision making has both been a focus of HCI research, such as on aesthetic design choices [17], and an outcome of research, such as in parental decision-making [34]. In effort to bolster this recent momentum, we formulate and explore the feasibility of a decision-support model that makes use of the complete **decision-making trifecta** (rationale, emotions, and intuition). It is our intention to reinvigorate the conversation of including intuition in decision making by utilizing

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our collective expertise. The first author of this paper is a software engineer with twenty years combined practice and teaching experience in mindfulness, yoga, meditation, breathwork, and other grounding practices that connect us to our deeper sense of self. The second author is an HCI and DSS researcher that maintains a regular yoga, meditation, and well-being practice.

One of our motivations for engaging in this work is the high amount of stress we see our students, colleagues, and industry partners experience. Stress comes from many directions, and we do not aim to cover those all in this paper, but we recognize that our work is often a major contributor. For example, in a qualitative study on the emotional wellbeing of HCI researchers engaging with sensitive topics, specifically those pertaining to End of Life, Moncur [41] validates the occurrence of researchers feeling stressed, distressed, and troubled by their work. In response to these findings, Moncur suggests that researchers consider a number of factors that could help them plan forward for their emotional wellbeing. In a recent study outlined in [44], the first author explored the benefits of specific mindfulness intervention programs on wellbeing, perceived productivity, and self-efficacy in the context of software engineering. A relevant finding was that participants were perceiving decisions as a cognitive burden given their everyday stress levels. We are motivated to explore the relationship between decreased decision making stress and Software Engineer and HCI practitioner wellbeing. One central value that we want to support in this work is bringing grounded, intentional decision-making to the fields of software engineering and HCI. We aim to bring mindfulness practices to the ethos of software development, because there is too much emphasis on gain, profit etc. and not enough questioning on whether a system should be developed [3]. We posit that bringing mindfulness to our work as software engineers and HCI researchers will encourage intentionality about the values we embed the systems we design. Furthermore, as Calvo and Peters' [8] claimed, "any move towards more ethical design and technologies that genuinely improve our lives requires that those technologies respect our psychological needs." Critically, as we explore the implementation of DSSs into mindfulness intervention programs, we want to avoid developing "just because [we can]" [36], especially as technology is often seen as reducing mindfulness [16].

In this paper, we present our exploratory research into an Intuition-Inclusive Decision Guidance Model, based on the Trifecta of Self-Enquiry, that we designed to support decision making independent of a specific context. We ground this model in the triangulation of data from (1) an analysis of existing theory, discussed in the background, (2) interviews with fourteen mindfulness and decision-making instructors across the world, and (3) a comparative analysis of open-ended decision-making applications. Before you continue reading, we invite you to take a deep breath and notice how it fills your body. Hold it at the top for a bit, then release with a sigh and relax into this moment. Stay present with the physical sensations in your body for a few seconds, then carry on reading.

2 BACKGROUND & RELATED WORK

The Connection of Feeling, Thought & Belief. Based on work in psychology [33], media culture [53], and affective computing [42],

we use the following terminology: An *affect* is a physical experience or visceral sensation that comes up in the body, and quickly "washes" through us [47, 53, 64]. A *feeling* is a "sensation that has been checked against previous experiences and labelled" [53]. An *emotion* is the projection or display of a feeling, and can transfer into opinions and beliefs [42]. Given a trigger, we experience affects as visceral sensations, something in our body. We interpret the affects into feelings (such as joy, sadness, shame, or contentment), which arise, peak, and then subside, usually within a time frame of about 90 seconds due to the involved neuro-chemicals in the body [57]. Thus, we connect the feeling to the experience we are having in that moment. That feeling can trigger another thought, which can in turn intensify that feeling. That intensification can create a feedback loop with more thoughts following, and this becomes an emotion, a longer lasting experiential state. Every time we follow a train of thoughts, a neuronal pathway gets reinforced, we reinforce the beliefs that the mind trains along with these emotions. Over time, these beliefs can become so strong, most often unconsciously, that they inform our actions and choices, and can introduce biases. This model is a basis for the trifecta proposed later on.

Mindfulness. Mindfulness, the state of awareness in the present moment, is a well-researched path for reducing the effects of perceived stress. "Imagine feeling calm and content wherever you are. Imagine feeling that your life is so complete - right here, right now - that you do not wish to be anywhere else." [9] And science has proven the worth of engaging in it: Mindfulness is beneficial in cognitive therapy [29, 30] and pain therapy [65, 66], for reducing stress reactivity [18, 39], loneliness [38], systemic low-grade inflammation [62], cellular aging [14], and addiction [7]. The practices have positive effects on eating behavior [15], increase positive emotions [37], memory [28], and resilience [27]. Mindfulness meditation training positively impacts the brain [10] and may help elderly people retain optimal brain function [19]. In educational environments, mindfulness approaches are successfully applied in caregiver training [51], for clinicians [52], and for software engineers [4]. We aspire to replicate these benefits in our work with computer workers and software engineers [44] and, specifically for the paper at hand, incorporate them into everyday decision-making processes.

Mindfulness in HCI. For HCI, Terzimehić et al. [58] provided a detailed review and analysis of mindfulness research just last year. Their thematic analysis found seven lines of research, amongst them Reflection and Knowledge Gain as well as Mindfulness in Daily Life. Reflection and Knowledge Gain regards mindfulness as a practice of self-reflection and recollection, often practiced informally using journals. Mindfulness in Daily Life focuses on achieving mindfulness in daily life contexts such as sleep, nutrition, or technology disengagement. Both of these lines are beautifully exemplified and elaborated on in Akama, Light and Bowen [1] around the application domains of gardening, photography and videography. In the paper at hand, we contribute to the thematic line of Reflection and Knowledge Gain by offering a model for reflecting on self-inquiry, and relate to the challenge of technology disengagement also identified by [59].

Decision-Making Theory. Since the 50s, psychologists have investigated the rational decision-making process, and Simon was

the first to limit the notion of rationality because of cognitive limitations [54]. To get beyond that boundary, Gigerenzer [21] explores heuristics that help people make decisions where they cannot know all the facts. They propose a set of *heuristics* as a rational toolbox. Tversky and Kahnemann [60] claim that the normative and the descriptive analyses of choice should be viewed separately; Kahnemann [31] shows that the heuristics of experience lead to *biases*. Naturalistic Decision Making theory instead relies on a repertoire of patterns to prime recognition [35]. For a descriptive theory of decision making under *stress*, Janis [26] presents a typology of five distinctive patterns of coping behavior (including vigilance, hyper-vigilance, and defensive avoidance). In contrast, “*mindful*” judgment and decision making research leverages our knowledge about psychological processes into causal explanations for important judgment and choice regularities, and thereby can help to overcome judgments [63]. Specifically, Ruedy and Schweizer have shown positive effects of mindfulness on ethical decision making [48]. When we manage a positive mindset in addition to mindfulness, our choices seem to become more sustainable - as Isen [25] presents the impact of mild positive affect on thinking and motivation to understand affect’s influence on decisions. While all of these works shed light on how humans make decisions from the rational mind, including mindfulness, little attention is given to actively integrating the deeper sources of knowing in the body. Interestingly though, Kahnemann & Tversky’s Prospect Theory framework [32, 61] lists five phenomena directing human decisions - (i) framing effects, (ii) risk aversion, (iii) non-linear preferences, (iv) loss aversion and (v) source dependency - and all five are reflected in our model (see Sec. 4.1).

Decision Support Systems. The second author has applied DSS theory to management of complex agricultural systems the last eight years, and with this experience we posit whether DSS theory can be applied to those complex life choices that keep us up at night. Generally, we moot whether deploying mindfulness techniques via a decision support tool platform could help users find clarity in those issues they’re going back and forth on. Could such a platform help users come to a decision on an immediate next step? DSSs provide interactive support to assist people in semi or unstructured decision making [22]. Based on Simon’s decision-making framework [55], a semi-structured decision is one in which one of the three stages of decision making (i.e., intelligence - problem identification/formation, design - establishment of potential choices/alternative solutions, and choice - when the decision is made) is unstructured [24]. Specifically, they address cognitive overload and support heuristic problem solving of complex problems [23]. Historically, DSSs were created to support managerial and business decisions [45], however, now researchers apply decision support system theory to anything from cover crop species selection in agriculture, as the second author is developing, to ethical analysis of work settings [40]. Now, virtually any tool that offers a user decision support, in contrast to absolving a user of decision making, and has a language system, presentation system, knowledge system, and problem processing system [6] is referred to as a decision support system [46].

When looking to the literature with this question of whether deploying mindfulness techniques via a decision support tool could

ID	Profession and/or Organization	Expertise (excerpt)
i1	Art of Living, non-profit organization	Guru, yoga teacher
i2	Mission Dharma Founder	Buddhist meditation teacher
i3	Writer & healing teacher	Breathwork, acupuncture, reiki
i4	Musician, speaker, meditation teacher	Meditation, yoga, activism
i5	Meditation & yoga teacher, coach	Pranayamic breathwork, meditation
i6	Yoga teacher, writer, artist	Yoga, art, aromatherapy
i7	Blissology Yoga Founder	Yoga teacher, ocean activist
i8	MAP Institute Founder	Psychotherapy, hypnotherapy
i9	Author, researcher	Embodiment coach, somatics, trauma
i10	Neurodynamic Breathwork Founder	Breathwork
i11	Blissology Yoga Founder	Ayurveda, yoga teacher
i12	Embodied Mindfulness School	Meditation & yoga teacher, coach
i13	The Breath Center Founder	Pranayama & Anatomy of Awakening
i14	Global Spirit Family Founder	Shamanism & Reiki

Figure 1: Interviewees for the Intuitive Decision Making Inquiry

help users find clarity in rumination, we found a history of consideration regarding the role of intuition and bias in decision making and thus decision support systems. Sauter argues for the inclusion of intuition heuristics to DSS models to facilitate choice among potential solutions [50]. Intuition, Sauter argued, can be used to contend with uncertainty and complexity when analytical information is insufficient or if inconsistent measure exist [50]. Pretorius’s [46] characteristics that define DSS addressing “wicked problems” include the discovery and removal of erroneous logic (i.e., bias). The discovery and removal of bias in rumination is central component of the intuition-inclusive decision guidance model we introduce in this paper. Unfortunately, we have not so far been able to locate any occurrence of applying DSS theory to everyday or personal problems within the literature, and are of the current perception that it has not been explored outside of organizational and professional contexts.

3 RESEARCH DESIGN & ANALYSIS

3.1 Research Design

The research methods applied in this study are a combination of interview study, comparative analysis, and explorative design research. We performed an **interview study** with fourteen experts on meditation, breathwork, mindfulness, yoga, where we carried out semi-structured interviews. The experts are listed in Tab. 1. The interviews were conducted in the course of recording a publicly available online summit and intended for education and distribution in the interested community. For the **comparative analysis**, we searched related literature on Google Scholar and the ACM Digital library, for apps we searched the App Store and Google Play, and for web tools we searched using Google and DuckDuckGo. This exploratory search does not claim to be exhaustive or encompassing, yet it yielded sufficient results to identify a clear gap. For the **explorative design research**, we built concept model addressing the research gap in iterative modeling sessions based on the identified themes and running it through user scenarios with examples.

3.2 Interview Analysis

The interviews were analyzed using thematic analysis and thick descriptions [20]. We analyzed the interviews before comparing existing tools to not unintentionally insert bias and cherry-picking themes to support our argument. The first author conducted the

interviews, took notes, later on rewatched parts of them, and transcribed portions identified as relevant to themes. We found six major themes around decision making, which denote human social action that describes not only a behavior but also the context [20] to show the complexity. Finally, we extracted quotes as evidence, see Table 1. How we get to embody certainty takes different forms, but central common themes that we identified across the interviews are: (1) Getting out of our head, (2) Getting into our body, (3) A differentiation of sources of knowledge (mind, heart and gut), (4) Trust and self-efficacy, (5) Being in a good state, and (6) Silence and turning inward.

Getting out of our head. There was a notion of letting the thinking mind rest, and unhooking from thought - by some interviewees even referred to as an addiction to the thinking mind. This comes up especially when we feel stuck around a decision. "The thinking mind is a great worker but can be a terrible master" [Tab. 1, i2]. It can be a debilitating experience to keep questioning a decision we have taken beyond trying to learn from the consequences, and it can be as debilitating to procrastinate some decisions until we are "certain" about them even though we have learned all we can at that specific point in time, because that certainty may never arrive.

Getting into our body. Whether it is simply spending time in nature, yoga, breathwork, shaking, walking, all interviewees agreed that getting into the body, connecting to our physical senses is important. All these practices activate the parasympathetic nervous system, which relaxes our flight or flight mode under stress, which in turn allows us to get to a higher level of consciousness than survival mode. For some, a specific type of movement is part of their daily practice, for others it is something they incorporate as they go [Tab. 1, i5, i7].

A differentiation of sources of knowledge. We noticed interviewees were referencing the brain (rational mind), the heart (emotions), and the gut (intuition) and the interaction between these. The rational **mind** will make up stories in the sense of confirmatory narratives that suit any decision we are leaning towards. For example, if I am trying to decide between two jobs that I am being offered, the way I talk about them will likely reflect an inner leaning towards one or another. Now the interesting part is that when we rely too much on rational arguments, we can always find counter arguments that keep us stuck in place. For example, if the one job offers less money but more career growth opportunities than the other one, then a friend may mirror back to us that we had been talking about wanting to grow. In turn, we may find ourselves then arguing that we are not sure we can maintain our quality of life and our lifestyle with the reduced salary and start arguing for the other job - and when the friend mirrors that, the mind will find another argument to make us lean into the other direction again. Such is the habit of the analyzing mind, to try and find arguments and counter arguments for anything. If the mind cannot compare and argue and keep itself busy this way, it will find another problem to ponder upon - if we let it. Our mind seeks to be perpetually engaged. We may think we want to stop thinking, but for many of us our minds continue chattering away unless we deliberately engage them with something else [56]. The **heart** stands for the feelings and emotions. As in Sec. 2, the term feelings for the more fleeting sensations (e.g., a bout of sadness or anger, and the term emotions for a more persistently perceived state (e.g., desire). Some people feel not very

connected to their emotions [Tab. 1, i12] while others base their decisions on it [Tab. 1, i3]. When someone says they decide based on how they feel, they may be acting on a spontaneous feeling experience, or on an emotion that they have developed around a certain topic. It gets a bit more difficult to differentiate that if we notice that beliefs we have created over time lead to certain emotions and we decide based on such an emotion that essentially relies on a (mentally perceived) belief that may stem from our personal upbringing, circumstances or habits. The **gut** or intuition is a source of great wisdom if we understand how to tune into it [Tab. 1, i4]. We experience a few major sensations more easily in the gut. One is fear, a primal instinct as a reaction to specific changes in our outside circumstances, or a response to a thought. Fear wants us to pay attention, to become cautious but also curious - it is a survival instinct, but surviving doesn't necessarily mean to run or back off like our primary instinct may want to hint. As we have evolved, the fear response gets triggered by unwanted change, by cognitive overload, and by stressful situations. In our current society, there is a lot of potential for any of these three happening without us really wanting to run away as a response. Becoming curious and leaning into the discomfort can teach us a lot about when we are trying to avoid change and maybe resists growing in the course of that. We may just need to take a step back, breathe deeply to process, and maybe rest for a bit. The other response that can come from the intuition, a positive response, is much more subtle and takes practice to notice and pay attention to, like a "whispering of the soul" [Tab. 1, i3] or an inherent Yes in the body [Tab. 1, i4].

Trust, confidence in the self, connection to self, self-efficacy Another component of good decision making is to develop trust in ourselves. Some interviewees expressed it as trust, others as confidence, some again as connection to self or self-efficacy. We take a decision and assure ourselves instead of letting the doubting mind question us because it always will. So the trouble is that the ego mind will dangle the carrot of wanting to be certain in front of us, but it is an illusion, as the certainty will come with commitment to a taken decision. We make decisions by embodying certainty before conditions [Tab. 1, i12]. There were several suggestions on how to develop that trust, for example by processing emotions through breathwork [Tab. 1, i10]. Several interviewees stated that they could feel truth in their bodies (i3: "You can feel truth. This is a skill all of us have. It's natural for us."), and that we all can develop a sensitivity for such a feeling in our bodies [Tab. 1, i5].

Being in a good state. Many interviewees emphasized the importance of cultivating a positive attitude, being in a state of contentment, or getting resourced and into a good state first before making decisions. When stressed out, we rarely make our best decisions, and our mind has an evolutionary engrained negativity bias [56].

Silence and turning inward. One more theme that was transparent in all interviews is the notion of presence, of silence, of subtleties that can only come through when we can listen, as the noise from the outer environment quiets down. This silence can be found in meditation, but also in simply taking a few deep breaths and turning our attention inwards, for example using the breath as a tether for attention.

There are two conclusions from the six identified themes: (1) they all require a mindfulness practice like yoga or breathwork to shift

Theme	Quote
Getting out of our head	i12: Resisting stuckness makes it worse, the 'should be somewhere else / should be further along' that our ego mind presents to us is a trick to keep us ruminating instead of releasing the mind and being content. So the gap between here and "where I should be" is suffering. I'd ask where in your body do you feel stuck? And then, what emotion is coming up for you with that? And then, what underlying belief is there about life? i2: Uncertainty is what life is, and maybe we can remind ourselves that we can't know.
Getting into our body	i5: For me it comes back to my breath. Taking deep breaths to exhale the tension and the energy of thinking as they are arising. From there just sit quietly and listen. If I am willing to soften my body, keep my exhales long, my mindstream slows down, and what is needed to be known arises, just maybe the next 1 or 2 bits. And if I am obedient and trusting to that, the guidance keeps coming. If I am in the neighborhood of worry and tension, I don't see much. i7: The secret for happiness is: the more I blur the line between where I end and where nature begins, the happier I am. So for me, in the yoga practice, especially as I spend so much time around the ocean, I want to make every movement a love letter to nature and to the ocean in particular.
Diff. sources of knowing - Mind	i2: I like to remind people that we have the capacity to be aware. Once I help somebody attune to being aware, and it's natural, we are born with it, and it's not well utilised. I follow-up with notice how you are aware and not lost in imagination, not imagining the past or future, but you are just with the experience you are having, there is more likely a quality of peace. i10: We all have that inner knowing, but our ego mind gets in the way, so we need to let go from it, and breathwork helps with it.
- Heart	i12: People are very disconnected from their hearts, from their emotions. We can develop anxiety around our own emotions. For example, there is anger arising, and then I judge myself for it, and then I feel shame. But I can't express that so I don't get agency over that and have resilience. Let's get to the heart, let's learn a new technology of connection to ourselves. What are you feeling? Are you connected to how you feel? How somebody answers that tells me a lot. i3: We all receive intuitive information in all of the ways. For me it is a clear feeling, knowing in my heart, it just arrives, and so it's been a process of when something arrives to be able to trust it and say "that landed, that feel true in my body". It matches the environment of my being. It feels right, it feels true. And then you experiment.
- Gut	i3: We can hold deep gratitude and still wish to grow. Any type of gratitude that asks you not to grow doesn't really make sense. If your soul is whispering at you, if there is a message that keeps coming through about leaving something (a place, relationship, course of study). It's important to listen to that. It's part of learning to trust our intuition. We can always make a different choice. i4: When you get that heavy feeling, your head is spinning, you are obsessively thinking about something, you have to ask everybody. That's not your intuition. Intuition comes and feels like a flash of insight, like joy, it comes and it feels like a weight has been lifted. Intuition is not that stuck feeling, and what so many of us do when we have that stuck feeling is we follow it and we think that's what we are supposed to do.
Trust & Confidence	i10: Uncertainty always creates fear and anxiety because the ego mind hates change and risk, so it'll kick in fear. When you try a new thing, you will experience anxiety. So even if change is healthy, your ego mind doesn't get that, it will try to hold us back. In breathwork you process through fear and see that you can handle all that stuff. There is an opportunity in everything that exists. And you just have to really connect and see what that opportunity is for you. i5: We can recognize what truth feels like. There are resonators in the body, for each of us individually. It is easy to know, whether it is how goosebumps move, or a subtle presence of feeling, or a warming of the center of the chest, or an inner Yes where I feel "that's true for right now". And I say for right now because tomorrow it could be different.
Good state	i12: You'll always find a problem if you look for it. Be happy now, for no reason. i8: So first get yourself to a good place, at peace and content. Then be in the flow, without push or pull, be in your body. Then open your mind up with the intuition to ponder without attachment.
Silence and turning inward.	i1: Everything is changing. There should be a reference point. Who you are, your spirit, your soul, that is your reference point that never changes. That is immortal. If you have a confusion in your life, if you have to take a big decision, close your eyes for 10-20 seconds because silence always brings clarity to your life. And silence is the mother of all creativity, of all innovation. When it's very noisy, you won't get clarity, when you relax, new ideas come, new thinking comes. Silence is the way to bring more clarity to our life, maybe meditation. i13: Gravity. Accept what is. Get clear on possibilities forward. Tolerance. It's all an inside job to get to harmony.

Table 1: Quotes by interviewees

our consciousness from survival mode to a more resourced level and (2) they align to form a foundation for a guiding framework for decision making. Next, we look at tools for individual decision making and are curious whether or how well they align with this guidance.

3.3 Comparative Analysis of Tools

We performed an ad-hoc analysis of applications and tools for supporting individual decision-making. Our primary search string was "decisions", and when that didn't yield sufficient results (in the general purpose search engines), we re-ran the search with "decision support" and "decision app". The search locations for apps were the App Store (iPhone) and Google Play (Android), and for web-based applications they were Google and DuckDuckGo. The types of decision making tools we found in both the AppStore and Google Play were: **Weighted multi-criteria decision analysis**. These were the most relevant tools that came out of the search and based purely on analytical thinking and rationale. **Pros and cons lists**. These were considered semi-relevant but are even more narrow than the first group. **Fortune wheels**. Out of scope but highly represented in the search results were fortune wheels that take the decision off your shoulders by taking it for you. **Stories**.

Out of scope but also highly present: Stories that leave the choice for a decision up to a user, many in a love or adventure context.

There were 50 results that came up for "decisions" on Google Play¹). There were many apps like the decision roulette and the story choices, but they are not considered relevant for our purpose. Running "intuition" yielded results for apps that claim to train the intuition, based on arcade game type engagement and predicting content before revealed, but nothing relevant for the purpose of the study at hand. In the AppStore, there were 93 results for the search string "decisions", and even more irrelevant results than in the Google Play search, including an app to compare movies, tool boxes for other apps, and many storytelling apps. "Intuition" yielded no useful results on a laptop, and on the phone it listed a number of intuition training apps, oracle card decks, games as well as many irrelevant results.

For web-based applications we searched "decisions", "decision support" and "decision app" on both the search engines Google and Duck Duck GO with hardly relevant results coming up on the first couple of pages. There were a number of business decision support apps that would provide more transparency and help make meetings more efficient, but only one result that would serve individual

¹Results spreadsheet: <https://tinyurl.com/y7w86fpr>

decision making as for the study at hand. If Kim were to use these tools, they would take a purely rational, analysis-based decision. Even if they are willing to enter feelings of passion into a multi-criterion matrix - how to assign the weight, and quantify emotions and desires in the face of numbers and so-called “facts” about safety and security? Thinking back to the interview results - that’s not yet the solution...

3.4 Summary of Analysis

The main insight and conclusion from the thematic analysis and the comparative analysis is that there is a significant gap. The main points the interviewees stated were in terms of **connecting with the body, tuning into intuition, and turning inward** to our inner sources of knowing. The majority of the tools we found are making a random decision for the user (randomizers), outsourcing the decision (polls and social), and analytical frameworks or processes for the rational mind. **None of the tools used the aforementioned themes in any way.** Could a tool (potentially) help a person such as Kim (see Sec. 1) in navigating their challenges? What would that tool be capable of?

Research Goal & Question: In the following, we are consequently exploring whether it is possible to drive a tool towards these themes at all. The identified *Research Questions* are: (1) What is a model that includes the three sources of knowledge for decision-making? (2) Is it feasible to implement technology support for the identified themes? (3) Is technology potentially hindering the connection as the answer is sought “outside”?

We are aware that the very introduction of a tool puts a step in between the person trying to come to a decision in their life and them grounding in their body and checking in with themselves as recommended by the experts. Therefore, one of the intentions is to keep a potential intervention simple and minimalistic as to not distract from the purpose and as a reminder that the tool is external, whereas the answer is really internal.

4 EXPLORATORY DESIGN: INCLUDING INTUITION IN DECISION GUIDANCE

4.1 The Trifecta of Self-Enquiry

Based on our own experience as instructor and practitioner and supported by the interviews, we developed a model for self-inquiry that helps to understand and thereby guide decision making, and tie it back into prospect theory [61]. We call the **trifecta of self-inquiry** the faculties inside that help us reflect upon our reality in different ways and that help us make decisions: Our mind (the analytical rationale), our heart (the emotions), and our gut (the instinct or intuition).

The **mind** can take on three different roles in this activity - either as idealogue, attorney, or observer. The *idealogue* role comes into play by default when we rationalize as rationalization is rooted in a foundation of beliefs that we hold based on cultural imprints, upbringing, education, and our life experiences. We are experiencing the idealogue role of our brain when we hear thoughts that reflect convictions and beliefs we hold when it comes to a decision, e.g. “I better study computer science instead of art because computer science is a profession with secure and stable jobs.” Note, we do not promote this belief. The *attorney* role is good at listing pros

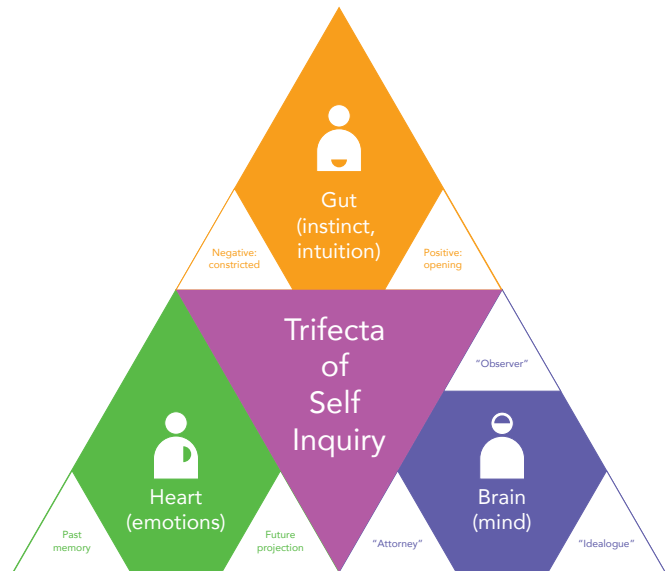


Figure 2: The trifecta of self-enquiry, design by Mafalda Samuelsson-Gamboa

and cons for either side of a decision. The attorney in us presents counter-arguments once we take sides and try to convince us of the other side. Thus while our rational in the attorney role is great at analyzing, it is less good at taking the actual decision. For example, when we think we have finally made up our mind we tend to hear that thought asking us, again, “but wouldn’t it be great to be a freelance artist and travel the world all the time and live uncaged?” Another indicator of is when we end up with the exact same number of items for each side of our pros and cons list. The *observer* role is more neutral and distanced to the stories that we tend to run in our thoughts, such that we can question assumptions and beliefs and instead become curious about the underlying stories and emotions. We are experiencing the observer role when we are watching our thoughts play out instead of being wrapped up in them, for example: “I am driven by my belief that I might not earn enough as an artist because I have seen freelance artist friends struggle to pay their rent and I fear that for myself” (i.e., reflecting the Prospect Theory phenomena of ‘framing’ and ‘source dependency’ discussed in the related work [61]). Truly, it hard to be aware of these roles within ourselves, particularly in the middle of a focused thought process, than it is to see them at work in Kim’s scenario.

The **heart** brings two main aspects to self-inquiry and decision-making, one that refers to the past, and one that infers to the future. Reflecting upon a choice, we involuntarily compare that choice to past situations. Emotions may come up that we have experienced during those past situations. Author Maya Angelou says we may forget what a person did or said, but we will remember how they made us feel [2]. Consequently, the emotions we have experienced in our past are likely to influence the decisions we make today, often reflecting the Prospect Theory phenomena of ‘non-linear preferences’ [61]. Exploring future scenarios in our imagination,

those projections are infused with emotions based on our past experiences. Consequently, as we project our future while we are planning ahead, these projections are based on how we have made our decisions in the past, how we have been brought up, in which environment we are in, and a whole list of socio-economic, cultural, and societal circumstances [11]. The engrained neuronal pathways from our past consequently lead to a highly predictable future where we repeat same outcomes and same feeling states - unless we diligently untangle ourselves from these patterns and train new feeling states. That means, if we desire a future that is different from our past, we need to train the feeling state we wish to reside in by deliberately evoking it in our bodies. In simple terms: If I want to move from unhappy to happy in my life, I need to train myself in feeling happiness instead of solely fixing an outer circumstance. By teaching ourselves and our physical bodies to experience the positive feeling state related to a desired future in the present moment, we move closer towards that future [11].

The **gut** will respond in one of two ways: negative in a constricting sensation, or positive in an opening sensation. The *negative* response is often some kind of fear. We can experience it as a constriction around our chest, a knot in our lower belly, a certain heaviness, or a jumpy reaction in our nervous system. Primal fear for survival is also triggered when we are exposed to stress hormones for an extended period of time. The primal fear trigger is why we don't make good decisions under high amounts of stress unless we specifically trained for that. Fear includes a perception of red flags that trigger protection mechanisms and instinct to avoid harm, becoming small, being caged/trapped, or losing agency (relating to the Prospect Theory phenomena of 'risk' and 'loss aversion' [61]). However, fear is also activated when we are asked to grow and go beyond our comfort zone for an opportunity. Our primal instincts do not like change, reinforced by evolutionary negativity bias so that risks are easily identified and prevented. If we can mindfully detect that our primal instincts have kicked in, we can make an informed choice about our decision and potentially decide to move beyond our comfort zone. The *positive* response is a sensation of opening or widening. We can experience it as a brightening sensation, a widening of the heart space, a weight being lifted, an opening of our visual space, or tingles of excitement. These positive reactions can be subtle, so it is good to experiment with noticing them in dedicated practice.² The gut is quickest to react, so when confronted with new information or a new decision we have to pay attention to detect the reaction before it gets overlaid by a story that our mind quickly infers about the new circumstance or a memory that gets triggered. Trickily, the mind wants to rationalize something as soon as it happens, so the mind's story can either confirm the sensation you just experienced, or it can present a counter-argument of why that initial hunch is not the right one. Thus, it is very important to notice the first physical, visceral reaction of unconscious knowledge. One way to train that is by practicing mindfulness.

²Take a couple of minutes to yourself, close your eyes, breathe deeply and calmly, and then ask your body: "Body, what does YES feel like?" and notice the sensations coming up. Then ask your body: "Body, what does NO feel like?" and notice the differing sensations. [12]

4.2 The Intuition-inclusive Decision Guidance Model

In this section, we introduce the training process and the continuous engagement that constitute the main parts of what we have named the "intuition-inclusive decision guidance model". The **training process** educates the user on the self-inquiry trifecta that was introduced in Sec. 4.1. The user learns how feelings, emotions, and beliefs come into existence and how to identify them. We walk through a running example with Kim, who was introduced in the previous section. This way, the user can relate to a set of three challenges that are common representatives for problems we tend to ruminate over. By relating their experience, they get a sense for how their own process may be similar to some extent, and how they may untangle old beliefs from actual constraints of a situation. When assessing beliefs, we have to take into account that the mind has an evolutionary negativity bias, so intuition felt in the body via embodiment practices can help to let go of limiting beliefs. We witness Kim exploring some of their held beliefs and how to work through them. To release emotions and beliefs and get a less entangled perspective, for example an old belief around money or love that says "I am not deserving", we invite the user to ask (1) whether that is true, (2) whether that is a helpful belief, and (3) whether they want to keep that belief. If not, here is the opportunity to replace it with a belief that serves better. Naturally that requires a lot of repetition of the new belief, e.g. in form of an affirmation and feeling into it, because if the body does not feel it, the mind won't believe it (detected to be a "lie" if there are contradicting signals in the body [13]) and will reject the new belief we are trying to train. The **continuous engagement** part invites the user to choose a particular challenge in their life they want to focus on in that session and explore what unconscious beliefs they may hold related to that. The user then gets walked through that same process and is encouraged to be present in their body and experience the sensations arising in the process. The user is invited to envision their decision and the feeling of discomfort that may arise with doubt around it. User tunes into that discomfort, to check what is underneath that discomfort and explore what gets triggered. We ask (1) What is the emotion that comes up? (2) Where do you feel that in your body? (3) And what beliefs are connected to that? [43] The user steps back into an observer role and takes the decision based on increased clarity on emotions and underlying beliefs. The user prepares for and acknowledges the arguing mind against whatever decision they made - and hopefully stands their ground.

The guidance model can be the core of a context specific DSS, but depending on an exploration of the implications of such an implementation, there could also be other ways to make it useful in educational or training contexts.

5 DISCUSSION

Implication not to design? Technology predominantly engages our visual sense and prefrontal cortex, our analytical mind is acutely focused, and we become disconnected from our body. In losing our connection to our body, we lose the felt sense of our gut, our intuition. Designing interactive technologies or advancing research on technology and mindfulness requires an unlearning of entrenched

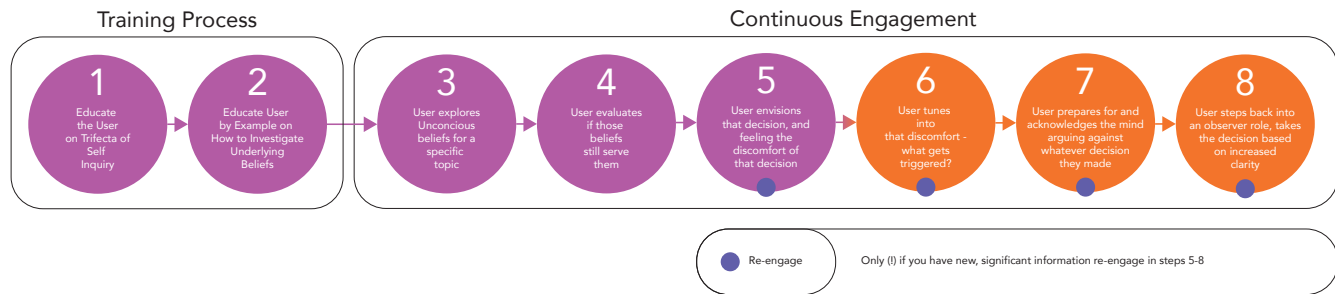


Figure 3: The Intuition-inclusive decision guidance model, design by Mafalda Samuelsson-Gamboa

constructs to fully immerse and re-engage with oneself and the world anew [1]. How do we unlearn the way how we previously engaged with technology to then explore how technology might support us differently? That's why we want to investigate this concept model.

Privilege. Not everyone is in the position make a decision like Kim in that scenario, and people with less privilege often have more complex problems that are harder to solve. We are not claiming all problems can be solved with a little more mindfulness or decision guidance. We are only offering a guidance model that can support decisions that are available for a person to take.

Beyond the scope of this paper. There are several topic areas well beyond the scope of this paper that we want to delimit from: (1) People who are not used to facing their emotions may be overwhelmed by a tool implementing the intuition-inclusive decision guidance model. (2) Mental health disorders require a medical professional, not an app that tries to navigate a person through a learning process. (3) If the decision one faces is inherently a wicked problem, or a moral dilemma, such that there is no straight-forward answer, our guidance model may not be sufficient to help [46].

Threats to validity. We identified four potential threats to validity for this research. There is a potential for researcher subjectivity in the interpretation of the interview data. The comparative analysis is not all-encompassing, but provided sufficient coverage for an overview of what is available. We constructed the trifecta model based on literature, interview analysis and experience, but is currently still disputable as it lacks validation by a psychologist. Our decision guidance model is a hypothesis grounded in analysis, but still untested. The last two points are avenues to future work.

Revisiting initial motivations. Early readers of this paper found the affect-to-belief formation and Trifecta of Self-Inquiry was empowering. Their take away? The opportunity to reflect makes a huge difference. We see similar indicators in the qualitative data collected in our breathwork study where we have used the trifecta in the accompanying guidance sessions³. The positive feedback we have received so far bolsters our intention to continue this line of work as we hope to support our peers and students.

6 CONCLUSION

In one sentence, this paper introduces an intuition-inclusive decision guidance model that is based on the hypothesis that we take

the most aligned decisions when our rational mind is being reigned in by a background check of the underlying emotions and beliefs we are running and when we are in touch with our intuition, which is based on a strong sense of presence in our physical body. Broken down, the contributions of this paper are as follows: We present why and when personal decisions can be challenging; We report on a thematic analysis of fourteen expert interviews on intuition and decision-making; We provide the results of a comparative analysis of tools and related work on this topic; We identify an evidenced research gap from the thematic analysis and the comparative analysis. The interview themes of connecting with the body, tuning into intuition, and turning inward were not considered at all in any of the tools we found. We propose the trifecta of self-inquiry and a decision guidance model for discussion that incorporates all themes identified in the interviews and the trifecta exemplified in a running example. We discuss the relevance of a potential tool implementation (or whether the implication is not to design one), the relevance of privilege in this context, problem types beyond the scope of this paper, and connected research domains and topics.

Future work. The intuition-inclusive decision guidance model could be implemented into a context specific DSS, however, we are not assuming that a DSS is the way forward. We have sketched mock-ups during our discussions of whether this challenge should and can be supported by technology, and we are in the process of refining further development for a tool prototype. We suspect there is room for DSS, and want to go through the exercise to find out, see [5] on questioning whether the technology is the solution and not just assume it is and design for it. We are collecting expert feedback, including by psychologists, and anticipate fruitful discussions around the decision guidance model and its realization in whichever form deems most useful.

Final words. Mental well-being is intimately tied to both social support and physical activity, so now go for a walk with a friend - and if we are still in a pandemic when you read it, go for a walk & call a friend while wearing a mask ;-)

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³<https://www.twinklflip.com/rise-2-flow/>

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